

Controllable Synthesis of a Well-defined Polypropylene grafted Silica Nanoparticles and Its Effect on Crystallization Behavior of Polypropylene

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Abstract Well-defined polypropylene grafted silica nanoparticles (PP-*g*-SiO₂) were prepared through the reaction of maleic anhydride grafted polypropylene (PP-*g*-MAH) with amino-functionalized silica (SiO₂-NH₂) by the 'grafting-to' method. The grafting density of PP-*g*-SiO₂ is found to be controlled by the concentration of silane coupling agent 3-[2-(2-aminoethylamino) ethyl amino] propyl trimethoxy silane (TAMS). The maximum grafting density of grafted PP-*g*-MAH chains with molecular weight of 9100 g/mol could reach 0.34 chains/nm², when the critical concentration of TAMS was 0.0194 mol/L. The critical concentration of TAMS can be explained by the maximum amounts of primary amino groups, which can totally react with PP-*g*-MAH on the surface of SiO₂-NH₂, when the silane monolayer is formed. The synthesized PP-*g*-SiO₂ with different molecular weights was mixed with PP by solution mixing to form a series of nanocomposites. The crystallization temperature (*T*_c) of nanocomposites increased significantly with the particle loading. The PP-*g*-SiO₂ with high molecular weight of grafted chains exhibits a high nucleation ability at 1 wt% nanoparticle loading in PP/PP-*g*-SiO₂ nanocomposites. In summary, we provide an effective method to synthesize the well-defined PP-*g*-SiO₂ with controlled grafting density, which shows excellent nucleation ability.

Keywords Polypropylene grafted silica nanoparticles; Grafting density; Grafting to; Crystallization behavior

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INTRODUCTION

Polymer nanocomposites (PNCs) have been extensively investigated in the past decades due to their excellent mechanical,^[1] thermal,^[2] flame retardant,^[3] electric,^[4] electromagnetic,^[5] optical,^[6] and gas permeability properties.^[7] Numerous works based on various polymer matrices have been reported by many researchers, of which polypropylene (PP) has received much attention owing to its light weight, low price and excellent processing performance.^[8,9] However, it remains a challenge to improve the compatibility between inorganic nanoparticle and organic polymer matrix in the preparation of PP/SiO₂ nanocomposites.^[10] One approach is the modification of silica or polypropylene matrix by silane coupling agents. Yang *et al.* conducted surface modification of SiO₂ by four different kinds of silane coupling agents (*n*-dodecyl-trimethoxy silane (DTMS), aminoethyl aminopropyl-trimethoxy silane (APTES), amino-propyl-trimethoxy silane (APS) and methacryloxypropyl-trimethoxy silane (MAPS)), then blended them with PP by melt blending.^[11] They found that the photo-stability and thermo-

oxidative stability of nanocomposites were related to the structures of silane coupling agents. Taniike *et al.* synthesized grafted-type polypropylene (silane-PP) with a few reactive groups per chain by direct copolymerization with the help of a Ziegler-Natta catalyst, and without damaging the stereoregular sequence of PP.^[12] Then the silane-PP as a reactive matrix was melt-blended with SiO₂ nanoparticles. Although this method could effectively increase the dispersion of SiO₂, the preparation process of silane-PP was complicated and the introduction of multiple reactive groups could decrease the crystallization rate and the Young's modulus of nanocomposites.

Because the single modification of silane has lots of limits, at present, the most promising approach to improve the dispersion of nanoparticles in the matrix PP is grafting polymer chains on the surface of nanoparticles. A series of non-PP chains were grafted on the surface of SiO₂ by 'grafting-to' method, such as polystyrene (PS), poly(methyl methacrylate) (PMMA), poly(ethyl acrylate) (PEA), and poly(butyl acrylate) (PBA).^[13–15] Then the grafted SiO₂ nanoparticles were blended with the matrix PP by melt blending method. Because of the low compatibility between non-PP-grafted chains and the matrix PP, the dispersion of nanoparticles in the matrix PP became worse and mechanical property of nanocomposite decreased as soon as the particle loading exceeded 0.8 (vol%).

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Compared with non-PP grafted chains,^[16] grafting PP chains onto the surface of SiO₂ can further increase the compatibility between SiO₂ and the matrix PP. Terano *et al.* synthesized polypropylene grafted silica (PP-*g*-SiO₂) from a well-defined terminally hydroxylated polypropylene (PP-*t*-OH).^[10,17–20] The grafted chains acted as the crystallization nuclei and co-crystallized with the matrix PP chains to make PP-*g*-SiO₂ act as the physical cross-linkage between lamellae. The dispersion of nanoparticles was still highly uniform until the particle loading reached 10 wt%. Although the performance improvement of PP/PP-*g*-SiO₂ is higher than that of PP/non-PP-*g*-SiO₂, the preparation of PP-*t*-OH is a challenge due to the low reaction temperature, low stereoregularity metallocene catalyst and complex pre-/post-treatment of PP-*g*-SiO₂.^[17]

According to the above discussion, the key problem is to find a simple and effective way to synthesize the well-defined PP-*g*-SiO₂. The commercially available maleic anhydride grafted polypropylene (PP-*g*-MAH) becomes a potential option, which is usually added into the PP/bare SiO₂ nanocomposites (as a physical compatibilizer) by melt-blending in order to increase their compatibility. Yang *et al.*^[21] and Bartczak *et al.*^[22] utilized reactive melt blending technique to prepare nanocomposites which consisted of amino-functionalized silica (SiO₂-NH₂, modified by silane coupling agents), PP-*g*-MAH and PP. Under melting condition, SiO₂-NH₂ reacted with PP-*g*-MAH by amidation to form covalent bond to increase the compatibility between SiO₂ and matrix PP. The weight loss of PP-*g*-SiO₂ synthesized by this process was about 80% in the range of 100–800 °C by thermogravimetric analysis (TGA), which may indicate that there were a great number of unreacted PP-*g*-MAH chains in PP-*g*-SiO₂. Therefore, this method synthesized a masterbatch of PP-*g*-SiO₂ and PP-*g*-MAH, which was difficult to calculate the accurate grafting density of PP-*g*-SiO₂. Wu *et al.* grafted three kinds of PP-*g*-MAH with different molecular weights onto the surface of amino-functionalized silica in tetrahydrofuran (THF) and xylene solution.^[23] Compared with neat PP, the Young's modulus of nanocomposites increased by 29%, but the yield stress and tensile strength increased less at 6 wt% particle loading, which might be attributed to the low grafting density.

In this work, we reported a method of synthesizing well-defined PP-*g*-SiO₂ in xylene solution by grafting commercially available PP-*g*-MAH chains onto the surface of SiO₂. The effect of concentration of silane coupling agent (3-[2-(2-aminoethylamino) ethyl amino] propyl trimethoxy silane, TAMS) (during amino modification) on the grafting density was investigated. It is worth noting that the maximum grafting density is 0.34 chains/nm², which is higher than previous literature,^[23] at the critical concentration of silane coupling agent TAMS (0.0194 mol/L). Meanwhile, the crystallization temperature (*T_c*) of PP/PP-*g*-SiO₂ prepared by solution mixing significantly increased at low particle loading, which indicated the high nucleation ability of the well-defined PP-*g*-SiO₂.

EXPERIMENTAL

Materials

Spherical silica nanoparticles with an average diameter of 60 nm were synthesized by means of Stöber method.^[24] The silane

coupling agent (3-[2-(2-aminoethylamino) ethyl amino] propyl-trimethoxy silane (TAMS)) and isotactic polypropylene with weight-average molecular weight (*M_w*) of 1.4×10⁴ g/mol (PP-14K) were purchased from MACKLIN. Maleic anhydride grafted polypropylene (PP-*g*-MAH) with *M_w* of 9.1×10³ g/mol and 9.0×10⁴ g/mol were obtained from Sigma-Aldrich and Eastman Co., Ltd., (Shanghai, China) with 8 wt%–10 wt% and 1 wt% maleic anhydride, respectively.

Aminosilane Coupling to Silica

Dry silica (2 g) was added into 80 mL of freshly prepared piranha solution (25% H₂O₂, 75% H₂SO₄) under magnetic stirring at 60 °C for 4 h. Then, the suspension was separated by centrifugation for 17 min at 12000 r/min. The obtained SiO₂ sediments were rinsed with deionized water for 0.5 h and ultrasonically dispersed for 7 min. This washing procedure was repeated five times until the pH of the washing water became neutral. Next, the washed SiO₂ was ultrasonically dispersed for 30 min and freeze-dried at −46 °C for 72 h to remove water. Finally, in order to further remove the absorbed water on the surface of silica, the SiO₂ was heated at 110 °C in a vacuum oven for 24 h before being used.

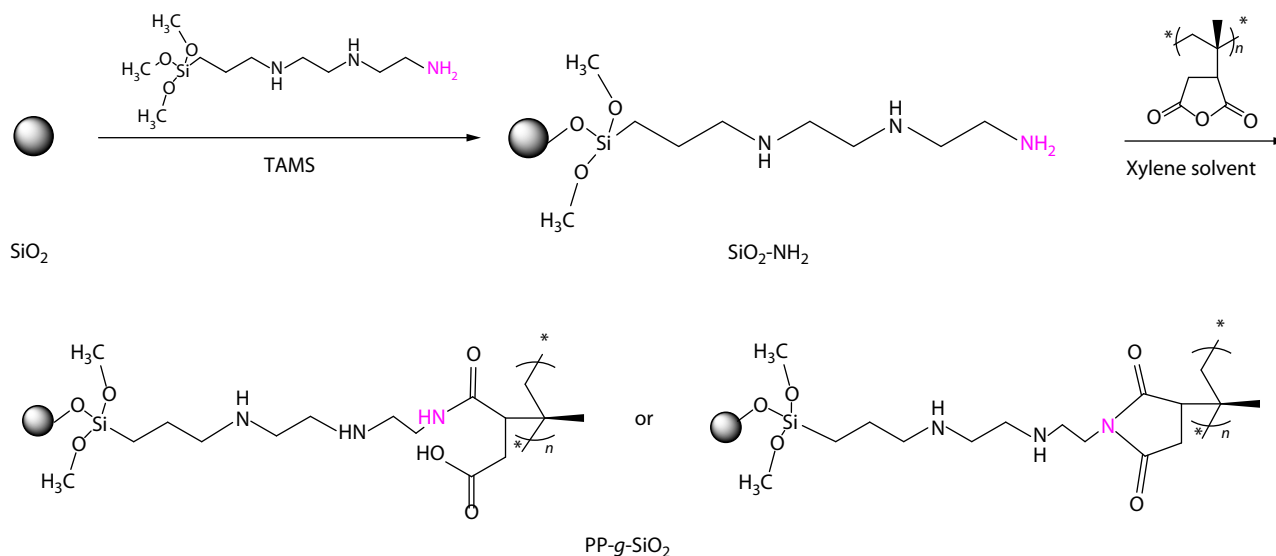
Salinization was carried out as follows. SiO₂ (125 mg) modified with piranha solution was added into a three-neck round bottomed flask with 12.5 mL of xylene. Then, the suspension was ultrasonically dispersed for 1 h at a power of 300 kW. Under nitrogen atmosphere, different amounts of TAMS were added dropwise into the suspension and refluxed at 115 °C for 12 h with constant stirring in an oil bath. After reaction and cooling to room temperature, the product was washed three times with fresh xylene and methanol to remove all of unreacted TAMS. Then, the washed SiO₂-NH₂ was dried in a vacuum oven at 25 °C for 12 h.

Grafting of PP-*g*-MAH onto Aminosilane Silica

The synthesis of PP-*g*-MAH-grafted silica (PP-*g*-SiO₂) nanoparticles by means of 'grafting-to' method is shown in Scheme 1. Both 50 mg of SiO₂-NH₂ and 30 mL of xylene were added into a 50 mL three-neck round bottomed flask, and the suspension was ultrasonically dispersed for 1 h at a power of 300 kW afterwards. Under the nitrogen protection, 100 mg of PP-*g*-MAH was added into the suspension and the flask was transferred into an oil bath. At the beginning, the suspension was refluxed at 120 °C for 0.5 h with constant stirring, then the heating temperature was changed to 100 °C with the continued reaction for 9 h. After reaction, the mixture was separated by centrifugation for 15 min at 12000 r/min. The product was dried in a vacuum oven for 12 h at 30 °C. Next, the dried powder was dialyzed by xylene for 2 h at 130 °C to remove all of the unreacted PP-*g*-MAH. Finally, the clean PP-*g*-SiO₂ was heated to 30 °C in a vacuum oven for 12 h in order to volatilize the residual xylene.

Preparation of PP/PP-*g*-SiO₂ Nanocomposites

To prepare the PP/PP-*g*-SiO₂ nanocomposites, PP and PP-*g*-SiO₂ were individually dispersed in xylene solvent and the SiO₂ solution was sonicated for 5 min before blending them in the desired weight ratios. The mixture was stirred for 5 h in an oil bath at 100 °C before casting onto Petri dishes. The nanocomposites were dried in a vacuum oven at 50 °C for 12 h to remove the solvent.



Scheme 1 Synthesis of PP-*g*-SiO₂ nanoparticles by 'grafting-to' method.

In this part, we used two kinds of PP-*g*-SiO₂ with PP-*g*-MAH chains of $M_w = 9\text{ k}$ and 90 k g/mol to blend with PP to form the nanocomposites. For clarity, the PP-*g*-MAH grafted SiO₂ was donated as xK-PP-*g*-SiO₂-y, where x and y represents the molecular weight of PP-*g*-MAH and grafting density, respectively. The characteristics of samples are summarized in Table 1.

Characterization

The ²⁹Si CP/MAS nuclear magnetic resonance (NMR) spectrum of dried SiO₂-NH₂ was obtained on a Bruker Avance III 400 solid NMR spectrometer which was operated at a resonance frequency of 79.3 MHz for ²⁹Si. The spectrum was recorded at a temperature of 25 °C, and the following conditions such as 4 mm, 5 kHz, 90° pulse, 5 ms contact time, 2 s relaxation delay were adopted.

In order to calculate the grafting density of PP-*g*-SiO₂ and test the modification effect of silane coupling agents, the weight losses of SiO₂-NH₂ and PP-*g*-SiO₂ in the range of 100–800 °C were measured by a thermogravimetric analysis (TGA, PE8000, USA) under nitrogen atmosphere and the gas flow rate was 20 mL/min. In general, about 5 mg of sample was heated from 50 °C to 100 °C at a rate of 40 °C/min and kept isothermal at 100 °C for 2 min to remove absorbed water on the surface of particles, then heated from 100 °C to 800 °C at a heating rate of 20 °C/min and the final weight of residual was also recorded.

Transmission electron microscope (TEM, HT-7700, Hitachi, Japan) at an operating voltage of 100 kV was used to observe

the dispersion and radius of nanoparticle. Firstly, a certain amount of PP-*g*-SiO₂ was added into xylene and ultrasonically dispersed for 1 h at room temperature. Then, the suspension was transferred into an oil bath and heated to 100 °C with constant stirring for 6 h. After stirring, two drops of hot suspension were placed on a TEM copper grid immediately and dried at room temperature for 0.5 h to volatilize xylene for further observation.

Fourier transform infrared (FTIR) spectroscopy measurements were performed by a Nicolet 6700 FTIR spectrometer with a MCT detector (Thermo Fisher Scientific, USA) under transmission mode. The collected range of spectra were 4000–650 cm^{−1} with a resolution of 4 cm^{−1}, and 32 scans were accumulated for each sample.

The surface chemical information of SiO₂-NH₂ and PP-*g*-SiO₂ was investigated on a Thermo Scientific ESCA Lab 250Xi with a 200 W Al K_α monochromator X-ray source through X-ray photoelectron spectroscopy (XPS) analysis. The base pressure of the analysis chamber was kept at about 3×10^{−10} mbar. The bonding energy of the main C1s component at 284.67 eV was used to correct the electron-bonding energy.

The crystallization behaviors of PP nanocomposites were studied on a differential scanning calorimetry (DSC, Q2000 TA Instruments) which was calibrated with indium. The samples (5–10 mg) were encapsulated in aluminum pans, and heated to 210 °C for 5 min to erase thermal history, followed by cooling to 0 °C and reheating to 210 °C. The first cooling and second melting thermograms were recorded. All tests were performed at a cooling/heating rate of 10 °C/min and used ultrapure nitrogen as purge gas.

The crystallinity (X_c) of nanocomposites was calculated by Eq. (1):^[21,25]

$$X_c (\%) = \frac{\Delta H_m}{\Delta H_m^0} \times 100\% \quad (1)$$

where ΔH_m is melting enthalpy normalized by mass fraction of PP in nanocomposites, ΔH_m^0 is the melting enthalpy of 100% crystalline PP (209 J/g).^[26]

Polarized light optical microscope (PLOM, Olympus BX51,

Table 1 The characteristics of samples for preparation of nanocomposites.

Sample	Weight-average molecular weight (g/mol)	Polydispersity index (PDI)	Grafting density (chain/nm ²)
PP-14K	1.4×10 ⁴	2.3	—
PP- <i>g</i> -MAH-9K	0.9×10 ⁴	2.4	—
PP- <i>g</i> -MAH-90K	9×10 ⁴	3.1	—
9K-PP- <i>g</i> -SiO ₂ -0.23	—	—	0.23
90K-PP- <i>g</i> -SiO ₂ -0.04	—	—	0.04

Japan) coupled with a camera system (Canon EOS80D, Japan) was used to observe the spherulitic morphology of PP-based nanocomposites. Firstly, the samples were heated to 210 °C and kept isothermal for 3 min to erase thermal history. Then, the samples were cooled to 125 °C at a rate of 20 °C/min. All photographs were taken after the crystallization was completed.

RESULTS AND DISCUSSION

Preparation of PP-*g*-SiO₂ by Grafting to Method

The surface chemistry of SiO₂-NH₂ and bare SiO₂ were provided by ²⁹Si CP/MAS NMR spectra. As shown in Fig. 1, both the bare SiO₂ and SiO₂-NH₂ have three resonances at −91.5, −101 and −110 ppm, which are named as S², S³ and S⁴, respectively. The S² species are related to the geminal silanol and/or methoxy groups. The S³ species are due to the isolated or vicinal single silanol groups and the single methoxy S⁴ species represent fully condensed silicate groups.^[27] After surface modification of SiO₂ by TAMS, a new peak at −66 ppm (P1, originated from aminopropyl groups)^[28,29] appears in the curve of SiO₂-NH₂, which indicates that TAMS have been successfully chemically bonded onto the surface of SiO₂-NH₂.

TEM was performed to observe the morphology of SiO₂ and detect the presence of grafted polymer layer on the surface of PP-*g*-SiO₂. As shown in Fig. 2(a), the average diameter of bare SiO₂ is about 60 nm with uniform dispersion and the surface of bare SiO₂ is smooth with a clearly visible boundary. After PP-*g*-MAH was grafted onto the surface of SiO₂ (Fig. 2b), a low-contrast layer can be observed around the particle and the boundaries of particles become obscured.

FTIR spectra of bare SiO₂ and SiO₂-NH₂ in the range of 3600–2800 cm^{−1} are shown in Fig. 3(a). For SiO₂-NH₂, the absorption bands at 2853 and 2926 cm^{−1} are attributed to the —CH₂— stretching vibrations from TAMS,^[30,31] which indicate that SiO₂ is successfully modified by TAMS. Fig. 3(b) shows the FTIR spectra of PP-*g*-MAH and PP-*g*-SiO₂ in the range of 1900–1500 cm^{−1}. For PP-*g*-MAH, the signal bands at 1844 and 1777 cm^{−1} can be assigned to the symmetric and asymmetric stretching vibrations of the carbonyl groups in cyclic anhydride, respectively. The band at 1712 cm^{−1} is assigned to the carbonyl groups of free acids after MAH hydrolysis. These re-

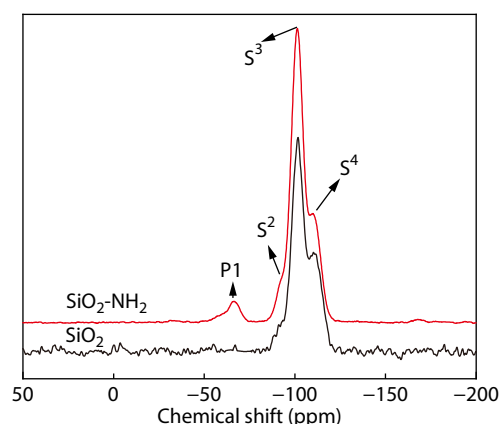


Fig. 1 ²⁹Si CP/MAS NMR spectra of SiO₂-NH₂ and bare SiO₂.

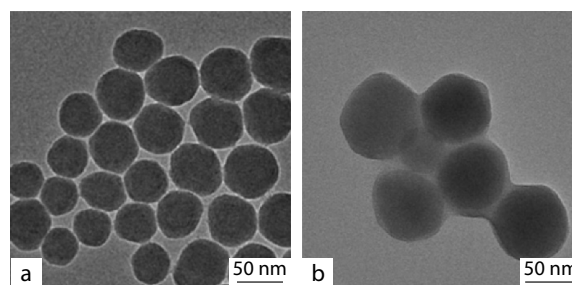


Fig. 2 TEM images of (a) bare SiO₂ and (b) PP-*g*-SiO₂.

sults indicate that the MAH groups in PP-*g*-MAH consist of two forms: cyclic anhydride and free acids.^[32] After PP-*g*-MAH was grafted onto the surface of SiO₂, the intensities of bands at 1844 and 1777 cm^{−1} significantly reduce or disappear. At the same time, new bands appear at 1697 and 1640 cm^{−1}, which are due to carbonyl groups of imide^[33] and stretching vibration of carbonyl groups in amide, respectively. Therefore, PP-*g*-MAH has been chemically grafted onto the surface of SiO₂-NH₂ via amidation reaction between anhydride and amino groups.

The surface chemical information of SiO₂-NH₂ and PP-*g*-SiO₂ was provided by XPS with high resolution scans. In XPS high resolution C 1s scan of SiO₂-NH₂ (Fig. 4a), the main peak at 285 eV is due to C—C bonds (C1) on the surface of SiO₂

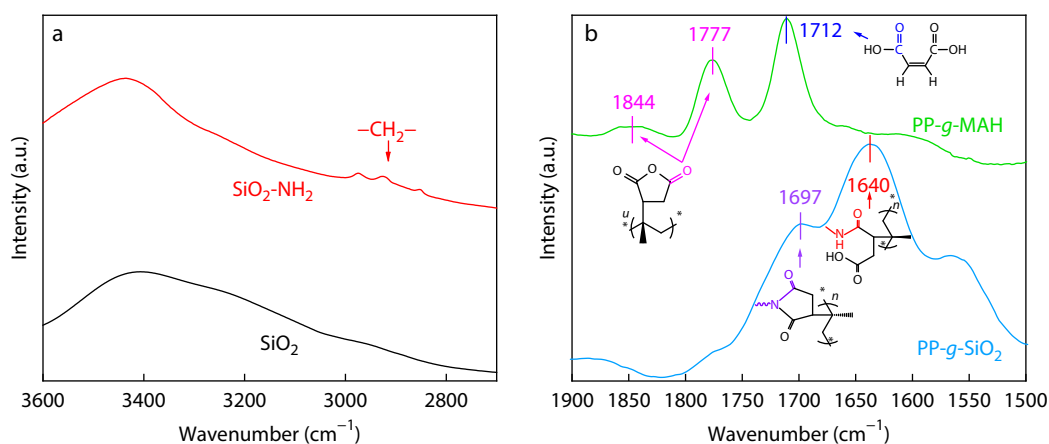


Fig. 3 Magnified FTIR spectra of (a) SiO₂ and SiO₂-NH₂, (b) PP-*g*-MAH and PP-*g*-SiO₂.

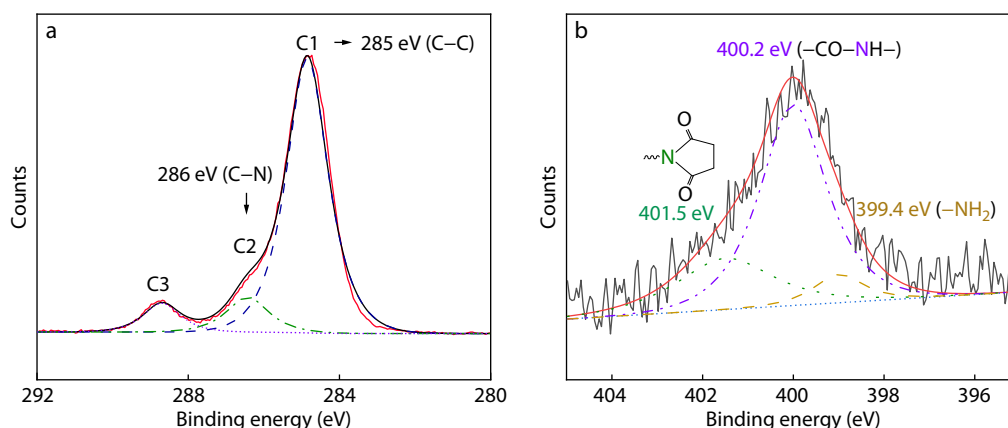


Fig. 4 XPS high resolution spectra of (a) $\text{SiO}_2\text{-NH}_2$ and (b) PP-g-SiO_2 .

nanoparticles. Meanwhile, the signal peak at 286 eV is attributed to C—N bonds (C2) from the amino groups of TAMS.^[34–36] Since there are no C—N bonds on the surface of bare SiO_2 , the appearance of C2 peak further proves the successful amino-modification of SiO_2 by TAMS. C3 peak is possibly ascribed to the presence of CO_2 .^[34] Fig. 4(b) shows the XPS high resolution nitrogen 1s scan of PP-g-SiO_2 , in which the signal peak at 401.5 eV is related to the formation of imide and the peak at 400.2 eV is attributed to the formation of amide.^[37] The appearance of these two peaks further suggests that PP-g-MAH has been successfully grafted onto the surface of $\text{SiO}_2\text{-NH}_2$ by amidation reaction between anhydride groups of PP-g-MAH and amino groups of $\text{SiO}_2\text{-NH}_2$. Finally, the peak at 399.4 eV is related to the unreacted -NH_2 on the surface of PP-g-SiO_2 .

TGA curves of bare SiO_2 , $\text{SiO}_2\text{-NH}_2$ and PP-g-SiO_2 were shown in Fig. 5. Thereinto, bare SiO_2 , $\text{SiO}_2\text{-NH}_2$, 9K- PP-g-SiO_2 -0.23 and 90K- PP-g-SiO_2 -0.04 display a weight loss of 4.8%, 10.6%, 16.5% and 21.4% from 100 °C to 800 °C, respectively. The TGA curve of $\text{SiO}_2\text{-NH}_2$ can be divided into three stages according to the decomposition temperature:^[38] the weight loss of 0.8 wt% at low temperature (<200 °C) is related to the removal of physically absorbed water on the surface of nanoparticles. As the temperature rises to 200–450 °C, the weight loss of 5.5 wt% is attributed to the condensation between the silanol groups on the surface of particles and the removal of by-product water. In the last temperature range (450–800 °C), the observed weight loss is attributed to the oxidation and de-hydroxylation processes of silane groups at high temperature. Subsequently, the grafting density of PP-g-MAH on the surface of nano-silica was calculated by Eqs. (2) and (3) as well as the weight loss of $\text{SiO}_2\text{-NH}_2$ and PP-g-SiO_2 :

$$G_r = \frac{W_{\text{Amino-silica}}^{\text{PP-g-silica}}}{1 - W_{\text{Amino-silica}}^{\text{PP-g-silica}}} \quad (2)$$

$$\text{Grafting density} = \frac{N_A \times G_r}{M_{\text{PP}} \times S_{\text{spe}}} \quad (3)$$

where $W_{\text{Amino-silica}}^{\text{PP-g-silica}}$ is the value of difference between the residual weight loss of $\text{SiO}_2\text{-NH}_2$ and PP-g-SiO_2 , N_A is Avogadro number, M_{PP} is the weight-average molecular weight of the PP-g-MAH , S_{spe} is the specific surface area of $\text{SiO}_2\text{-NH}_2$.^[39]

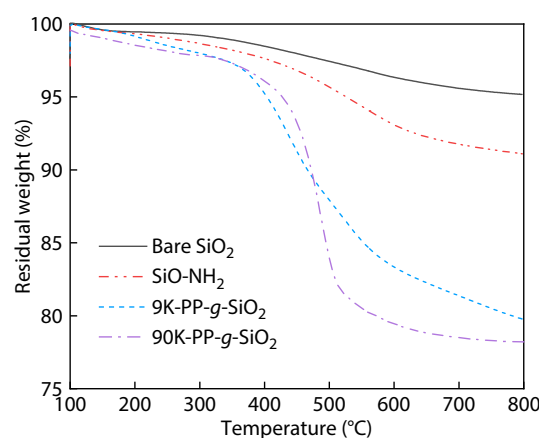


Fig. 5 TGA curves of bare SiO_2 , $\text{SiO}_2\text{-NH}_2$ and PP-g-SiO_2 .

Fig. 6(a) shows the weight loss curves of 9K- PP-g-SiO_2 -y modified with various concentrations of silane coupling agents. The plot of finally calculated grafting density of 9K- PP-g-SiO_2 -y based on Eq. (3) versus the concentrations of silane coupling agents was shown in Fig. 6(b). As shown in Fig. 6(b), the grafting density of 9K- PP-g-SiO_2 -y first increases rapidly, reaches a maximum value and then decreases gradually with the concentration of TAMS. Based on Eq. (3), the maximum grafting density at the critical concentration (0.0194 mol/L) is 0.34 chains/ nm^2 . This shows that changing the concentration of TAMS is an effective method to control the grafting density of PP-g-SiO_2 . Wu *et al.* ever grafted PP-g-MAH onto the surface of SiO_2 via similar process.^[23] After the calculation of TGA results, the grafting density of PP-g-MAH with the molecular weight of 2K, 8K and 30K was 0.13, 0.04 and 0.01 chains/ nm^2 , respectively, which was obviously lower than that of our work. The ratio of SiO_2 to 3-aminopropyl-dimethylethoxysilane in their reactant during amino modification was about 15–19. During the amino modification process in our work, when the concentration of silane coupling agents was 0.0019 mol/L (the minimum concentration of TAMS), the ratio of SiO_2 to TAMS was about 20 which was comparable to the value of Wu *et al.* At this point the grafting density of PP-g-SiO_2 was 0.18 chains/ nm^2 , which was also comparable to the maximum grafting density of Wu *et al.* So

the concentration of silane coupling agents was too low to provide an adequate amount of primary amino groups, which might limit the increase in the grafting density of Wu *et al.*

The change of grafting density with the concentration of silane coupling agents can be explained by the reaction between alkoxy-silanes groups of TAMS and the silanol

groups on the surface of $\text{SiO}_2\text{-NH}_2$ during amino modification. The structural evolution of TAMS on the surface of $\text{SiO}_2\text{-NH}_2$ shown in Fig. 7 contains three stages. In the first stage (Fig. 7a), when the concentration of TAMS is very low, some isolated TAMS molecules are located onto the surface of SiO_2 . With the increase of the concentration, the isolated TAMS

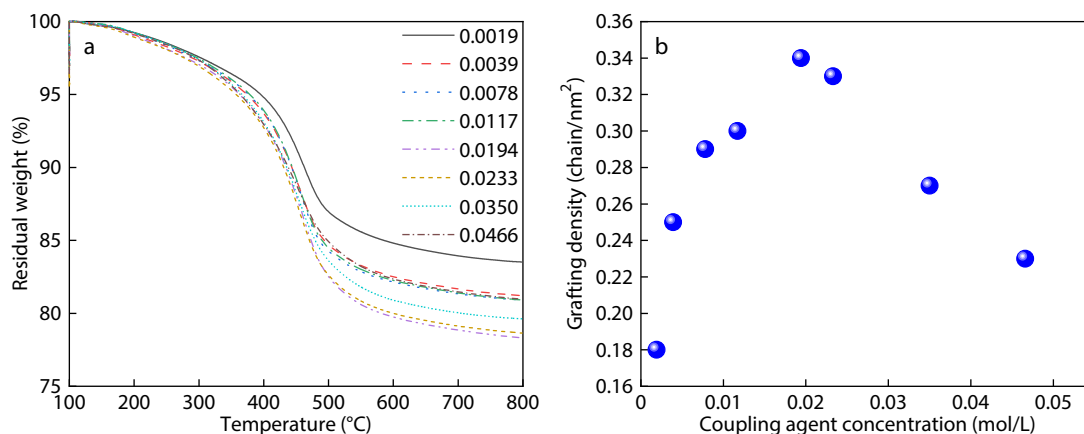


Fig. 6 (a) TGA curves of 9K-PP-g-SiO₂-y modified with different concentrations of TAMS; (b) Plot of grafting density versus the concentrations of TAMS.

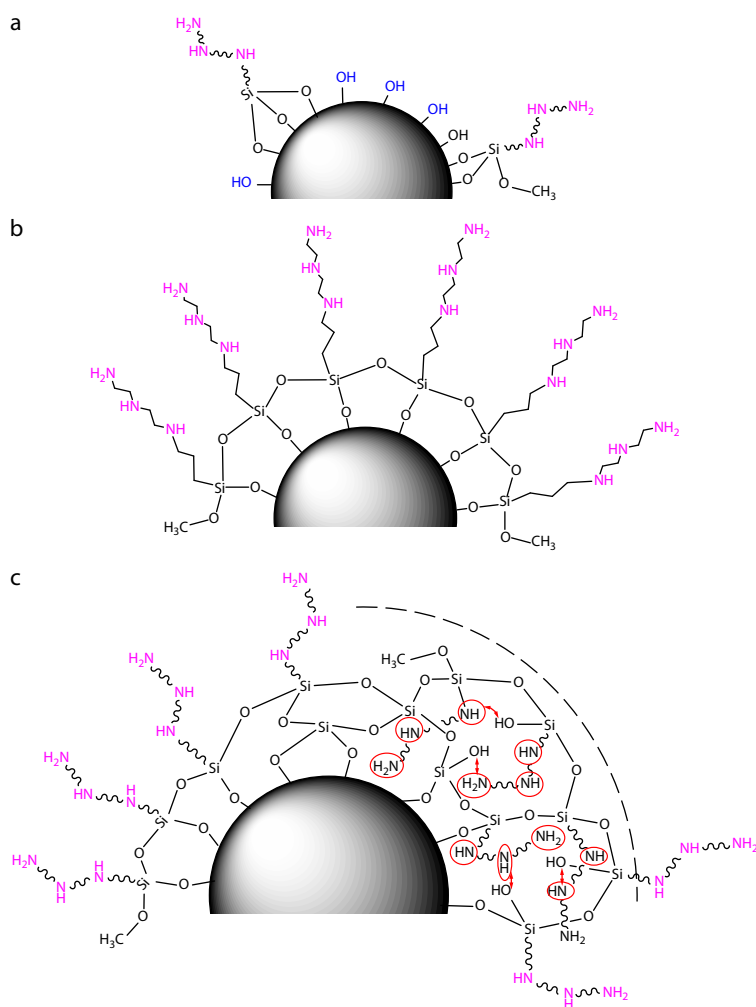


Fig. 7 Schematic illustration of the structural evolution of TAMS on the surface of $\text{SiO}_2\text{-NH}_2$.

molecules begin to link with each other through Si—O—Si bonds. The silane monolayer^[40–42] (the second stage, Fig. 7b) is formed until the concentration of silane coupling agents reach a critical value (0.0194 mol/L). In these two stages, almost every amino group in the TAMS can be served as reaction sites (primary amino groups marked by pink fonts in Figs. 7a and 7b) and react with PP-g-MAH totally. Therefore, the amounts of primary amino groups and grafting density both increase with the concentration of silane coupling agents until the silane monolayer is formed.

When the concentration of TAMS exceeds the critical value (the third stage, Fig. 7c), tri-functional alkoxy-silanes could self-condense and form local cross-linked structures in the presence of a trace amount of water from xylene solvent^[43–45]

(the area marked by the black dotted line in Fig. 7c) except for the reaction with silanol groups on the surface of SiO₂. Once these local cross-linked structures are formed, the amino groups on the surface of silica are divided into two types: one kind of primary amino groups (marked by pink fonts in Fig. 7c) located on the surface of multilayers will act as reaction sites and react with PP-g-MAH; On the contrary, the other amino groups (marked by red circle in Fig. 7c) lose the ability to undergo a grafting reaction, since they are shielded by silane cross-linked structures or hydrogen bonds^[46,47]. With the increase of TAMS concentration, local silane cross-linked structures are formed at an increasing number of sites,^[48] with the result that most amino groups are shielded by silane coating. Therefore, the primary amino groups and the graft-

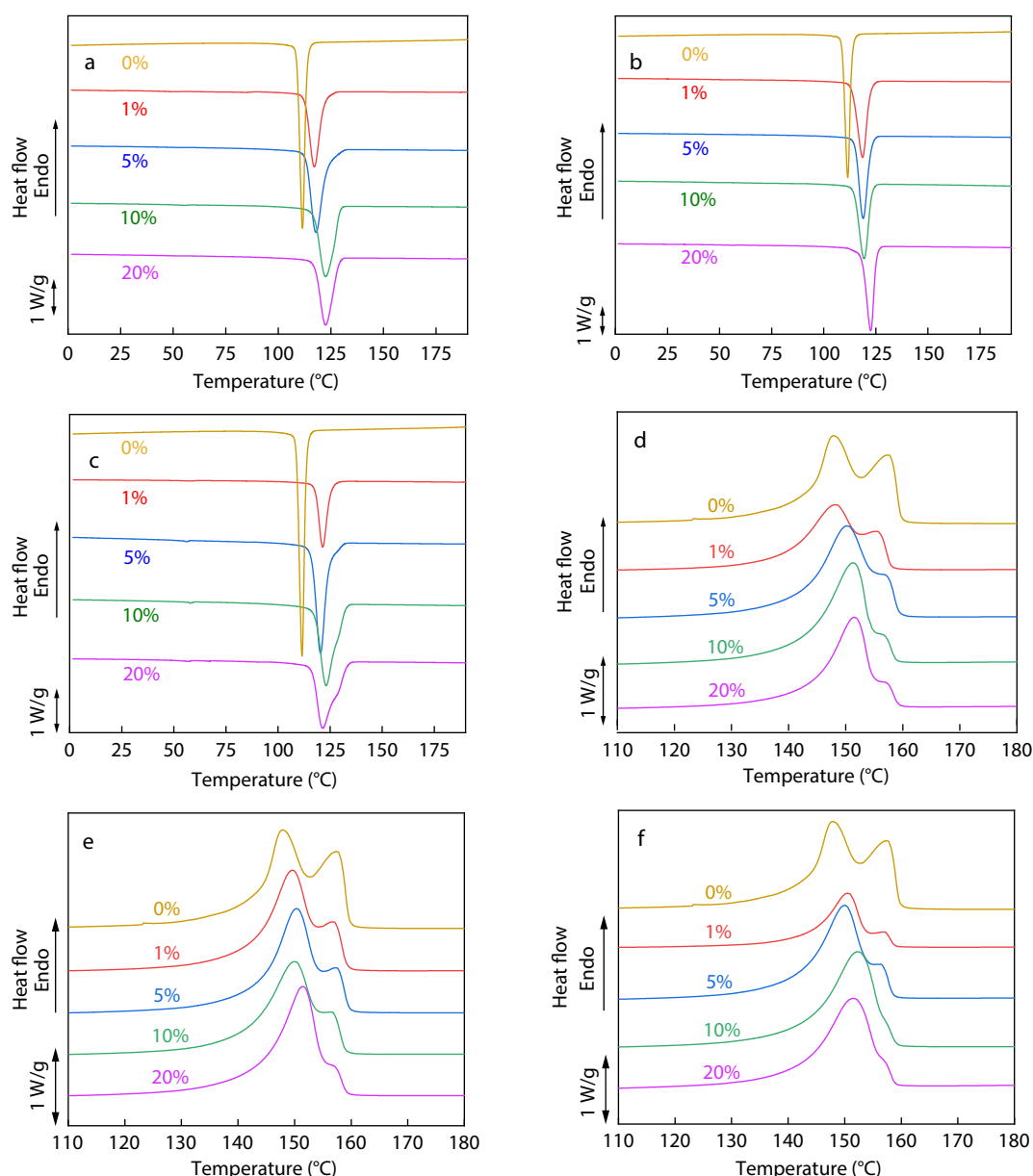


Fig. 8 DSC cooling curves of (a) PP-14K/bare SiO₂, (b) PP-14K/9K-PP-g-SiO₂-0.23, (c) PP-14K/90K-PP-g-SiO₂-0.04 with different particle loadings; The second heating curves of (d) PP-14K/Bare SiO₂, (e) PP-14K/9K-PP-g-SiO₂-0.23 and (f) PP-14K/90K-PP-g-SiO₂-0.04 with different particle loadings.

ing density both decrease with the concentration of TAMS in this stage.

Crystallization Behaviors of PP/PP-*g*-SiO₂ Nanocomposites

DSC cooling curves of PP-14K based nanocomposites with different kinds of silica nanoparticles and particle loadings are shown in Figs. 8(a)–8(c). Compared with neat PP, the crystallization temperature (T_c) of these three kinds of nanocomposites increases significantly with the addition of nanoparticles, which indicates the effective nucleating agent of all three kinds of silica nanoparticles. When the content of SiO₂ is in the range of 1 wt%–5 wt%, the T_c of nanocomposites is in the order 90K-PP-*g*-SiO₂-0.04 > 9K-PP-*g*-SiO₂-0.23 > bare SiO₂, in which the molecular weight of grafted chain PP-*g*-MAH plays an important role. Particularly, the longer grafted chains in PP-14K/90K-PP-*g*-SiO₂-0.04 nanocomposites provide a great number of heterogeneous nucleating sites which make T_c rapidly reach the higher value (121.4 °C) at 1 wt% particle loading as shown in Fig. 8(c). As for melting behaviors (Figs. 8d–8f), there are two melting peaks which named as low melting peak (LMP) and high melting peak (HMP) in all three kinds of nanocomposites. The LMP in neat PP is attributed to the melting of the smaller, less-perfect crystals. These imperfect crystals will reorganize or recrystallize during the melting scan, which results in the occurrence of HMP.^[49,50] The addition of PP-*g*-SiO₂ particles may retard the mobility of PP chains and

suppress the recrystallization,^[51] which improve the stability of crystals and result in the slight increase in the temperature of LMP and decrease of HMP area in nanocomposites.

The crystallinity and T_c of PP-14K based-nanocomposites with different kinds of silica nanoparticles and particle loadings are summarized in Fig. 9. As shown in Fig. 9(a), the T_c increase of PP-14K/90K-PP-*g*-SiO₂-0.04 nanocomposites at 1 wt% particle loading is about 10 °C which is obviously higher than other reported PP/SiO₂ nanocomposites as summarized in Table 2. If the T_c value of PP containing 1% of additive increases 6.5 °C or more compared with neat PP, the nucleation efficiency of additive can be rated as high.^[52] This further proves the high nucleation abilities of the well-defined PP-*g*-SiO₂. Fig. 9(b) shows that the crystallinity is almost unchanged in all three kinds of nanocomposites, which was also observed in other PP-based nanocomposites.^[53,54]

The PLOM results of PP-14K based nanocomposites with different kinds of silica nanoparticles and particle loadings are summarized in Fig. 10, which were used to evaluate the effect of silica types or particle loadings on the spherulitic morphology of nanocomposites. With the increase of particle loading, the nucleation densities of nanocomposites significantly increase and the sizes of spherulites decrease simultaneously, which indicate the nucleating abilities of all three kinds of silica nanoparticles. Specially, the nucleation density of PP-14K/90K-PP-*g*-SiO₂-0.04 is higher than the other two systems at the same particle loading, which can be explained

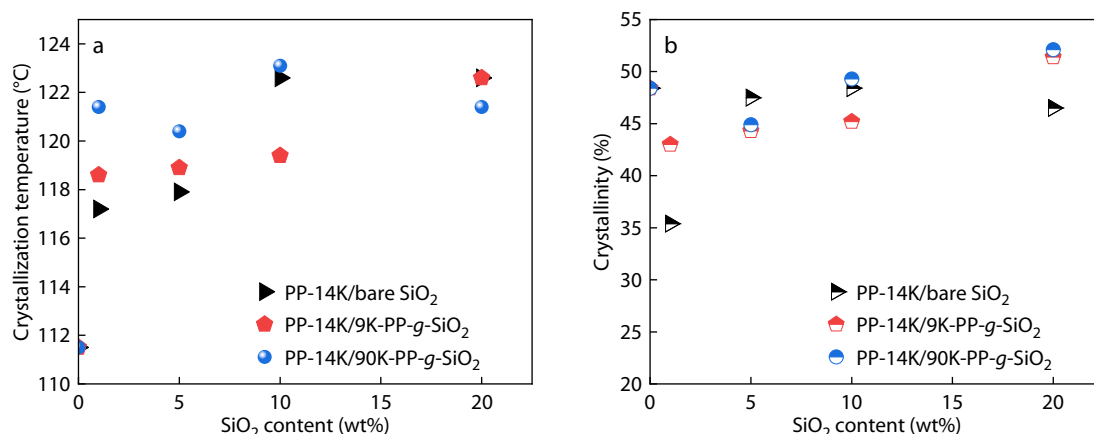


Fig. 9 (a) Crystallization temperature and (b) Crystallinity of PP-14K based-nanocomposites with different silica contents.

Table 2 The T_c increment of PP-based nanocomposites in previous studies.

Sample	Filler content (Φ_x , wt%)	T_c of neat PP (°C)	T_c of nanocomposite at Φ_x (°C)	Ref.
PP-OTMS5/SiO ₂	5.0	122.0	127.0	[12]
PP- <i>g</i> -VTEOS/oct-SiO ₂	5.0	119.4	118.3	[55]
PP/HBP- <i>g</i> -SiO ₂	5.0	117.8	118.1	[56]
PP/PP- <i>g</i> -MAH/SiO ₂	15.0	115.0	122.0	[57]
PP/S-1	1.0	113.7	122.9	[21]
PP/PP- <i>g</i> -MAH/am-SiO ₂ (S50)	1.25	120.7	127.4	[22]
PP/PP- <i>g</i> -SiO ₂ -2K	5.0	117.7	121.8	[23]

PP-OTMS5: PP copolymer with 15 mmol of (7-octen-1-yl) trimethoxysilane; PP-*g*-VTOS: vinyltriethoxysilane (VTEOS) modified polypropylene; Oct-SiO₂: octylsilane modified SiO₂; HBP-*g*-SiO₂: aliphatic hyperbranched polyester grafted SiO₂; S-1: a PP-*g*-SiO₂ masterbatch with 1 wt% of SiO₂; am-SiO₂: (3-aminopropyl) triethoxysilane modified SiO₂.

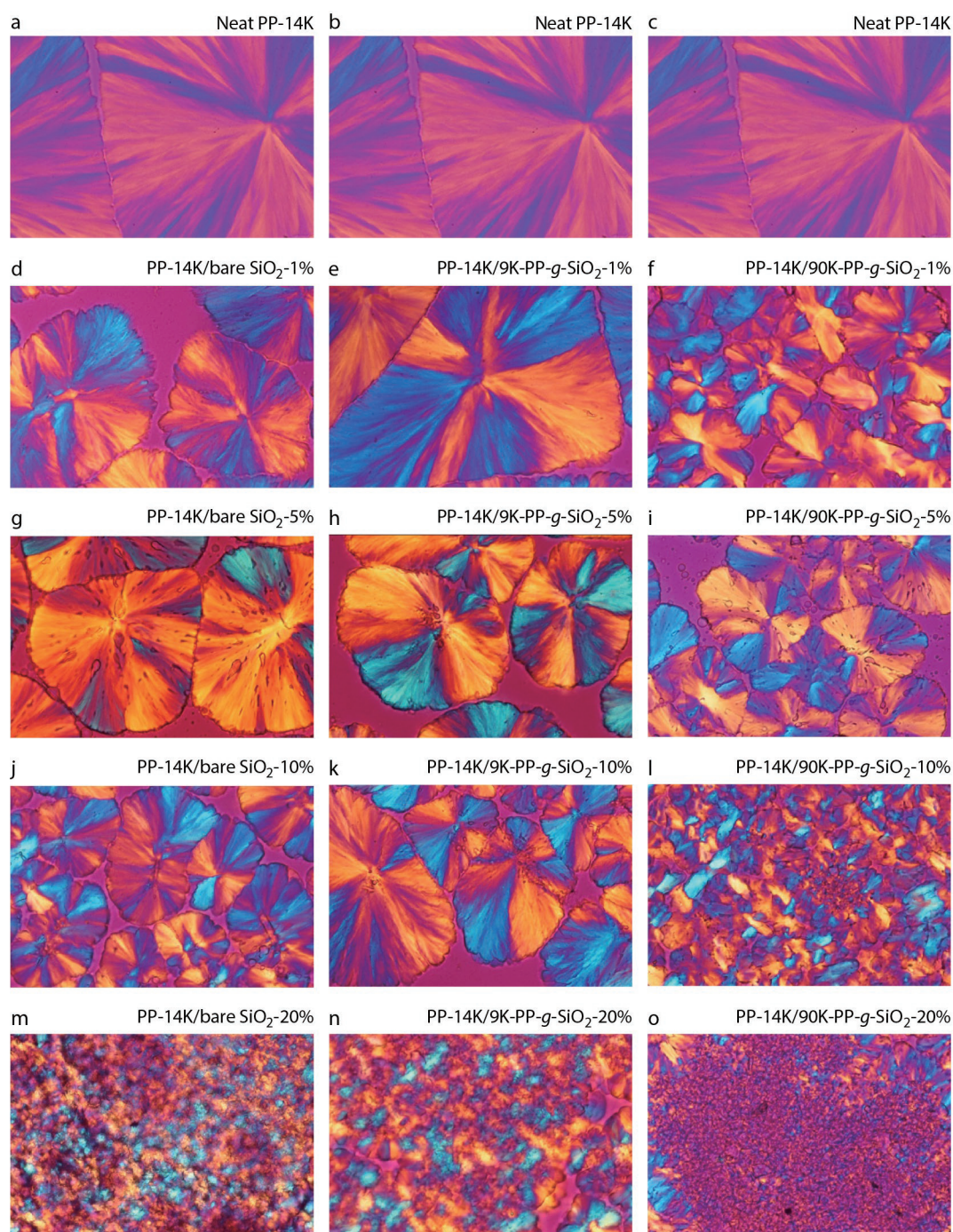


Fig. 10 PLOM results of PP-14K based nanocomposites with different kinds of silica nanoparticles and particle loadings.

by the larger number of heterogeneous nucleating sites provided by longer grafted chains in PP-14K/90K-PP-*g*-SiO₂-0.04 nanocomposites. The changing tendency of nucleation density with silica types and particle loadings is consistent with the non-isothermal DSC results.

CONCLUSIONS

In this work, the well-defined PP-*g*-SiO₂ by 'grafting-to' method was synthesized in xylene solution and the grafting density

could be controlled by the concentration of silane coupling agent TAMS. The maximum grafting density can be reached when silane groups on the surface of SiO₂-NH₂ could form a monolayer by Si—O—Si bonds with the increasing concentration of TAMS. At this point, every primary amino group in the TAMS can be served as reaction sites and react with PP-*g*-MAH totally. This work provides a method not only to synthesize the well-defined PP-*g*-SiO₂ by grafting commercially available PP-*g*-MAH onto the surface of SiO₂-NH₂, but also to simply and efficiently control the grafting density of PP-*g*-SiO₂

by regulating the concentration of TAMS. Meanwhile, the rapid increase of T_c in PP/PP-*g*-SiO₂ nanocomposites at low particle loading indicates that the well-defined PP-*g*-SiO₂ with longer grafted chains has greater nucleation efficiency.

NOTES

The authors declare no competing financial interest.

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